

Predicting Healthy Life Expectancy Using Machine Learning with Environmental, Nutritional, and Health System Indicators

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Abstract—Accurately predicting Healthy Life Expectancy (HALE) is essential for identifying populations at risk and allocating healthcare resources effectively. While previous machine learning studies have successfully predicted life expectancy using socio-economic and clinical indicators, they overlook upstream determinants such as environmental, nutritional, and health-system factors. This study extends an existing HALE prediction framework by integrating PM_{2.5} air pollution exposure, prevalence of undernourishment, and current health expenditure into a feature set of eight original indicators. Using country-level data from 185 countries spanning 2000–2019, we trained three machine learning models—Linear Regression, Random Forest, and Gradient Boosting—under both baseline (8 features) and extended (11 features) configurations, employing a temporal train-test split to simulate real-world deployment conditions. Gradient Boosting achieved the best performance with an R^2 of 0.9799 and RMSE of 0.9097 years, demonstrating that ensemble methods effectively capture nonlinear relationships between health determinants and HALE at age 60. The ablation study revealed that while Gradient Boosting benefits from the additional variables (RMSE reduction from 1.34 to 0.91 years), Linear Regression and Random Forest showed slight performance degradation, indicating that the upstream determinants carry modest but structured predictive signal best captured by boosting-based methods. These findings support the integration of environmental and nutritional indicators into HALE frameworks, aligning with recommendations from the Global Burden of Disease Study to adopt more comprehensive health prediction models.

I. INTRODUCTION

Life expectancy has long served as one of the most widely used indicators of population health and socio-economic development. However, a longer life does not necessarily imply a healthier one. As populations age, a growing proportion of additional years gained may be spent in states of illness, disability, or dependency [1]. This distinction has led researchers and policymakers to adopt Healthy Life Expectancy (HALE) as a complementary metric that quantifies the number of years a person can expect to live in good health. Unlike conventional life expectancy, HALE accounts for years lived with disability, making it a more informative measure for health policy planning, resource allocation, and the evaluation of public health interventions.

Accurately predicting HALE at the country level is valuable for several reasons. It enables governments and international organisations to identify populations at risk of spending a disproportionate share of their remaining years in poor health, to allocate healthcare resources more effectively, and to evaluate the potential impact of policy changes on population well-being. The Global Burden of Disease (GBD) 2017 study [1] demonstrated that while global life expectancy at birth increased by 7.4 years between 1990 and 2017, HALE at birth increased by only 6.3 years, indicating that not all gains in longevity translated into gains in health. Such findings underscore the need for predictive models that can identify the key determinants of healthy ageing across diverse national contexts.

Recent studies have applied machine learning techniques to predict life expectancy and related health metrics using socio-economic and clinical indicators sourced from the World Health Organization (WHO) and World Bank datasets [2] [3]. Ensemble methods such as Random Forest and Gradient Boosting have consistently outperformed linear models in capturing the complex, non-linear relationships between health determinants and longevity outcomes. Pranjali et al. [7] extended this line of work by developing a machine learning framework specifically targeting HALE at age 60 (HALE₆₀) using eight WHO health indicators across 185 countries from 2000 to 2019. Their study achieved strong predictive performance, with Random Forest attaining an R^2 of 0.9943. However, the authors noted that the feature set was limited to socio-economic and clinical variables and recommended that future work incorporate air quality, dietary patterns, and access to health services to enrich the predictive framework.

This study builds upon the framework proposed by Pranjali et al. [7] by integrating three additional variables that address the identified gap: PM_{2.5} air pollution mean annual exposure ($\mu\text{g}/\text{m}^3$) [4] as an environmental indicator, prevalence of undernourishment (% of population) [5] as a nutritional indicator, and current health expenditure (% of GDP) [6] as a proxy for health system capacity. These variables are sourced from the World Bank’s World Development Indicators and

will be merged with the original WHO dataset to construct an extended feature set of eleven predictors. Three machine learning models (Linear Regression, Random Forest, and Gradient Boosting) are trained and evaluated under both baseline (eight features) and extended (eleven features) configurations, with performance compared using RMSE and R^2 to quantify the contribution of the newly added upstream health determinants.

The remainder of this paper is organised as follows. Section II reviews the related literature and identifies the gaps addressed by this study. Section III describes the methodology, including the base framework, the proposed extension, dataset preparation, and the evaluation strategy. Section IV will present the experimental results and discussion, and Section V will conclude the paper with a summary of findings and directions for future work.

II. RELATED WORK

Research on healthy life expectancy (HALE) prediction spans two broad themes: epidemiological assessments of global disease burden and data-driven predictive modelling using machine learning. This section reviews key contributions in each theme, identifies recurring limitations, and explains how the present study addresses them.

A. Global Disease Burden and HALE Trends

The Global Burden of Disease (GBD) 2017 study [1] assessed disability-adjusted life-years (DALYs) for 359 diseases across 195 countries from 1990 to 2017. A central finding was that global HALE at birth increased by 6.3 years over the period, consistently lagging behind the 7.4-year gain in overall life expectancy, which implies that a substantial share of the additional years was spent in poor health. Disparities were pronounced: in high Socio-demographic Index (SDI) countries 26.3% of extra life years fell into poor health, whereas in low-middle SDI countries the figure was 11.7%. The study also documented a shift in the global disease profile, with age-standardised DALY rates for communicable diseases declining by 41.3% while total DALYs from non-communicable diseases rose by 40.1%. Although the GBD framework reports HALE at birth, the patterns it reveals, particularly the growing burden of non-communicable disease at older ages, are directly relevant to HALE at age 60, which is the prediction target of the present study. The GBD authors recommended that maximising HALE requires greater attention to upstream determinants, including environmental and dietary risk factors.

B. Machine Learning for Life Expectancy Prediction

Several studies have applied machine learning to predict life expectancy at the population level. Kumari and Siddiqui [2] trained linear regression ($R^2 = 0.96$) and decision tree models on behavioural and socio-economic predictors such as smoking, alcohol consumption, exercise, and healthcare access. Their feature set, however, did not include environmental, nutritional, or health-system indicators, and the prediction target was general life expectancy rather than HALE. The

authors recommended incorporating environmental variables in future work.

Sharma et al. [3] evaluated a broader range of algorithms and achieved an adjusted R^2 of 0.9729 with Extra Tree Regression, identifying GDP per capita, resource composition, and education as key predictors. While confirming the suitability of ensemble methods, their study similarly omitted ambient air pollution, dietary adequacy, and healthcare expenditure, and did not target health-adjusted outcomes.

Lipesa et al. [8] applied XGBoost, Random Forest, and Artificial Neural Network regressors to WHO data from 193 countries, with XGBoost achieving the lowest RMSE of 2.402. Their analysis highlighted infant mortality, schooling, and regional location as important predictors. Although the study incorporated a wider set of health and behavioural variables than [2] and [3], it did not include air quality, nutritional adequacy, or health expenditure indicators, and the prediction target remained general life expectancy.

Across these studies, ensemble methods consistently outperform linear models, yet the feature sets remain centred on socio-economic and clinical variables. None of the three studies targeted HALE or incorporated the upstream determinants discussed below.

C. Upstream Determinants of Health

Beyond the socio-economic and clinical indicators used in the studies reviewed above, three upstream determinants have been independently linked to population health outcomes in the epidemiological literature. In the environmental domain, Apte et al. [9] demonstrated that ambient PM_{2.5} exposure significantly reduces life expectancy at both global and regional scales, establishing air quality as a critical risk factor for cardiovascular and chronic respiratory disease. In the nutritional domain, Fosso Djoumessi [10] showed that malnutrition remains a primary constraint on infant mortality and life expectancy in Africa, underscoring the importance of dietary adequacy as a population-level health determinant. At the health-system level, Morina et al. [11] found that higher health expenditure is positively associated with life expectancy, confirming current health expenditure as a meaningful proxy for access to health services. Despite this evidence, these three dimensions, environmental exposure, nutritional adequacy, and health system capacity, have not been jointly incorporated into machine learning frameworks for predicting HALE.

D. Identified Gaps and Positioning of the Present Study

Two limitations recur across the reviewed literature. First, existing machine learning studies predict general life expectancy and do not distinguish between years lived in good health and years lived with disability. Second, feature sets have concentrated on socio-economic and clinical variables while underrepresenting the environmental, nutritional, and health-system dimensions established in Section II-C. Both the GBD 2017 study [1] and Kumari and Siddiqui [2] have explicitly

called for the inclusion of such upstream determinants in future predictive frameworks.

This study extends the HALE prediction framework proposed by Pranjali et al. [7] by integrating PM2.5 mean annual exposure ($\mu\text{g}/\text{m}^3$) [4] as an environmental variable, prevalence of undernourishment (% of population) [5] as a nutritional variable, and current health expenditure (% of GDP) [6] as a proxy for health system capacity into the existing WHO Global Health Observatory feature set used to predict Healthy Life Expectancy at age 60 (HALE_60). To quantify the contribution of these additional features, models are trained both with and without the new variables, and performance is compared using RMSE and R^2 , enabling a direct assessment of the predictive gain attributable to the inclusion of upstream health determinants.

III. METHODOLOGY

A. Base Framework

Pranjali et al. [7] developed a machine learning framework to predict Healthy Life Expectancy at age 60 (HALE_60) using country-level health data. The dataset was sourced from the World Health Organization (WHO) Global Health Observatory (GHO) and covered 185 countries over a 20-year period (2000–2019), producing 3,700 country-year observations. The target variable, HALE_60, represents the expected number of healthy years remaining for a person aged 60. Eight predictor variables were used, all socio-economic or clinical in nature: alcohol consumption per capita, probability of dying from chronic diseases between ages 30 and 70, mean total cholesterol, prevalence of physical inactivity, obesity prevalence, prevalence of raised blood pressure, tuberculosis deaths excluding HIV, and adult mortality rate.

Missing values were handled through a two-tiered median imputation strategy. Missing entries were first imputed using the country-level median across years, preserving localised temporal trends. Any remaining gaps were then filled using the global median of the respective indicator. Features were normalised using standard scaling for symmetric distributions and log transformation followed by standard scaling for skewed variables such as tuberculosis deaths and adult mortality rate. Seven machine learning models were trained and compared: Linear Regression, Ridge Regression, Lasso Regression, Random Forest, Gradient Boosting, AdaBoost, and Support Vector Regression. Models were evaluated using Root Mean Square Error (RMSE), the coefficient of determination (R^2), and cross-validation RMSE. After hyperparameter tuning, Random Forest achieved the best performance with a test RMSE of 0.55 and R^2 of 0.9943, followed by Gradient Boosting with a test RMSE of 0.58 and R^2 of 0.9937. Adult mortality rate and physical inactivity were identified as the two most influential predictors of HALE at age 60.

B. Proposed Extension

This study extends the framework of Pranjali et al. [7] in two ways: expanding the feature set with upstream health determinants, and improving several methodological aspects of

the preprocessing and evaluation pipeline. The original framework relied entirely on socio-economic and clinical indicators—variables that capture the consequences of poor health, such as disease prevalence, mortality, and obesity, rather than the upstream conditions that shape those outcomes. In their future work section, the authors stated that the framework could be improved by “including more variables in the form of air quality, dietary patterns, and access to health services, which are expected to enrich the predicting framework” [7].

This study directly addresses that recommendation by introducing three dimensions that were entirely absent from the original model: (i) an environmental dimension, captured through ambient air pollution exposure, which is a well-documented risk factor for cardiovascular and chronic respiratory disease; (ii) a nutritional dimension, captured through population-level dietary adequacy, which is a fundamental constraint on health outcomes particularly in low-income settings; and (iii) a health-system dimension, captured through national healthcare expenditure, which reflects a country’s capacity to prevent and treat disease. The specific indicators, data sources, and preprocessing procedures used to operationalise these dimensions are described in Section III-C.

In addition to the feature expansion, several preprocessing improvements were implemented to address methodological limitations identified in the original pipeline. These include replacing the random train-test split with a temporal split to prevent look-ahead bias, applying feature scaling after the split to avoid data leakage, keeping the target variable (HALE_60) in its original unit for interpretable error metrics, and adding RMSE and MAE alongside MSE as evaluation metrics. These changes are detailed in the subsections below.

C. Dataset and Preprocessing

The dataset used in this study builds upon the original WHO GHO panel described in Section III-A by incorporating three additional country-level indicators from the World Bank’s World Development Indicators (WDI). PM2.5 mean annual exposure ($\mu\text{g}/\text{m}^3$), sourced from the Global Burden of Disease Study 2023 via the World Bank [4], represents ambient air pollution as an environmental risk factor. Prevalence of undernourishment (% of population), obtained from the FAO via the World Bank [5], reflects nutritional deficiency at the population level. Current health expenditure (% of GDP), sourced from the WHO Global Health Expenditure Database via the World Bank [6], serves as a proxy for healthcare system capacity and access. Together, these expand the predictor set from eight to eleven features.

All three indicators are available for the same 185 countries and the same 2000–2019 period as the original dataset, ensuring full alignment. The WDI variables will first be transformed from wide format into a country-year longitudinal structure and then merged with the WHO data using country and year as composite keys. Since all countries in the original WHO dataset are also present in the World Bank sources, no records will be excluded during merging, preserving the completeness of the final panel at 3,700 observations.

TABLE I
PREDICTOR VARIABLES AND SOURCES

#	Variable	Description	Source	Status
1	Alcohol consumption per capita (litres)	Average annual alcohol consumption among individuals aged 15+, measured in litres of pure alcohol	WHO GHO	Original
2	Probability of dying from chronic diseases (%)	Probability of dying between ages 30 and 70 from cardiovascular diseases, cancer, diabetes, or chronic respiratory diseases	WHO GHO	Original
3	Mean total cholesterol (mmol/L)	Average total blood cholesterol level in the population	WHO GHO	Original
4	Prevalence of physical inactivity (%)	Percentage of adults aged 18+ not meeting recommended physical activity levels	WHO GHO	Original
5	Prevalence of obesity, BMI ≥ 30 (%)	Percentage of the population with BMI ≥ 30	WHO GHO	Original
6	Prevalence of raised blood pressure (%)	Adults aged 30–79 with systolic ≥ 140 or diastolic ≥ 90	WHO GHO	Original
7	Tuberculosis deaths (excl. HIV)	Number of TB deaths excluding HIV cases	WHO GHO	Original
8	Adult mortality rate	Probability of dying between ages 15–60 per 1,000 population	WHO GHO	Original
9	PM2.5 mean annual exposure ($\mu\text{g}/\text{m}^3$)	Population-weighted exposure to air pollution	World Bank WDI	New
10	Prevalence of undernourishment (%)	Insufficient caloric intake in population	World Bank WDI	New
11	Current health expenditure (% of GDP)	Health spending as percentage of GDP	World Bank WDI	New

Note: Green-shaded rows indicate the three variables added in this study. Target variable: HALE at age 60 (HALE_60), sourced from WHO GHO.

The feature set used in this study is organised into two categories. The first category comprises eleven core health indicators: the original eight from [7] plus the three newly added World Bank variables. These are the substantive predictors whose contribution to HALE_60 prediction is the focus of this study. The second category comprises engineered features derived from existing columns to improve model performance: HALE at birth (HALE_Birth), which is retained as an input feature because it captures a country’s overall health profile and is not a direct transformation of HALE_60; cyclical encodings of the year variable (sine and cosine) to represent temporal patterns; and one-hot encoded categorical variables (Location, ParentLocation, Dim1). The distinction between these two categories is important: the ablation comparison described in Section III-D operates on the eleven core health indicators, while the engineered features remain constant across all experimental configurations.

The inclusion of HALE_Birth as an input feature warrants justification. HALE at birth reflects population-wide health conditions from birth onward, whereas HALE_60 specifically measures expected healthy years remaining at age 60. The two metrics are related but not interchangeable, as HALE_60 is shaped by age-specific morbidity patterns at older ages that are not fully captured by HALE_Birth. However, because HALE_Birth exhibits high feature importance in preliminary experiments (see Section IV-D), results will be reported both with and without HALE_Birth to ensure that the contribution

TABLE II
FEATURE SET SUMMARY

Category	Variables	Count
Original health indicators [7]	Alcohol consumption, chronic disease mortality (30–70), cholesterol, physical inactivity, obesity, raised blood pressure, TB deaths, adult mortality rate	8
New health indicators (this study)	PM2.5 exposure [4], undernourishment [5], health expenditure [6]	3
Engineered features	HALE_Birth, year (sin/cos), one-hot encoded location variables	Variable

of the eleven core health indicators can be assessed independently. The missing value strategy in this study departs from the two-tiered median imputation used by Pranjal et al. [7] in order to better exploit the temporal structure of the panel data. The following three-step procedure is applied. First, for the year 2000, where no prior data exists, missing values are filled using the earliest available observation for the same country (backward fill). Second, remaining gaps between observed years are filled using linear interpolation within each country, estimating intermediate values from surrounding time points. Third, any residual missing values are imputed using the column-level mean as a final fallback. This approach preserves within-country temporal trends more effectively than median imputation alone, particularly for indicators that

Fig. 1. Methodology Overview

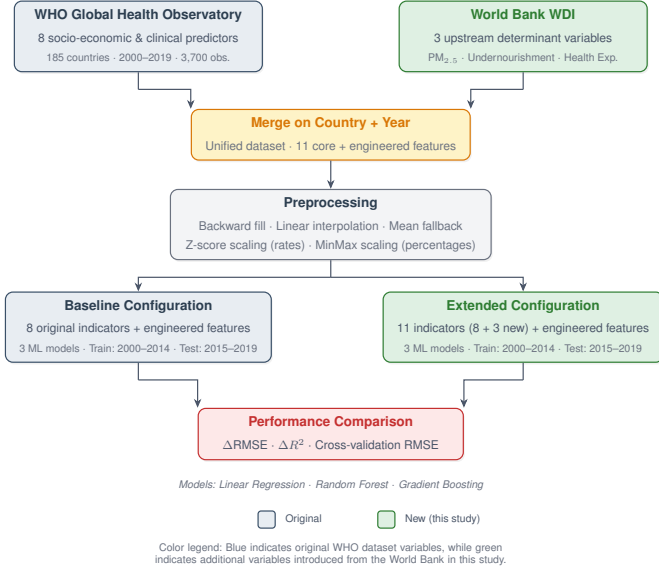


Fig. 1. Methodology Overview

exhibit gradual change over time. Columns with more than 60% missing values are dropped prior to imputation to remove uninformative features and reduce noise. The target variable, HALE₆₀, is kept in its original unit (years) and is not scaled, ensuring that prediction errors (RMSE, MAE) remain directly interpretable. For input features, scaling is applied after the temporal train-test split (described in Section III-D) to prevent data leakage. Numeric features representing rates and exposure levels are standardised using z-score scaling (StandardScaler), while percentage-based features are scaled to the 0–1 range using MinMaxScaler. The scaler is fitted on the training data only and then applied to both training and test sets. Figure 1 presents the overall methodological pipeline of the study, starting from data collection and merging of WHO and World Bank datasets through preprocessing, model training, and evaluation, while Table I summarises all predictor variables used in the study, clearly distinguishing between the original WHO indicators and the newly introduced World Bank features to ensure transparency in feature construction.

D. Evaluation Strategy

All models will be evaluated using Root Mean Square Error (RMSE), the coefficient of determination (R²), Mean Absolute Error (MAE), and cross-validation RMSE, consistent with the original framework [7] with the addition of RMSE and MAE for improved interpretability. RMSE and MAE express error in the same unit as HALE₆₀ (years), while R² indicates the proportion of variance explained. Cross-validation RMSE will be used to assess model stability and generalisation, with the standard deviation across folds indicating robustness.

To preserve chronological order and avoid look-ahead bias, a temporal split will be employed rather than random sampling. Data from 2000 to 2014 will be used for training,

while data from 2015 to 2019 will be reserved for testing. This replaces the random split used in the original framework and ensures that models are evaluated on genuinely future observations. In addition, 5-fold cross-validation will be applied during hyperparameter tuning to further evaluate model stability and reduce the risk of overfitting.

Three machine learning models were trained and evaluated: Linear Regression as a baseline, Random Forest as a tree-based ensemble, and Gradient Boosting as an advanced boosting method.

To isolate the contribution of the three newly added health indicators, each model is trained under two configurations: a baseline using the original eight core health indicators plus engineered features, and an extended configuration using all eleven core health indicators plus the same engineered features. Performance will be compared using the difference in test RMSE, defined as $\Delta RMSE = RMSE_{baseline} - RMSE_{extended}$, and the difference in test R², defined as $\Delta R^2 = R^2_{extended} - R^2_{baseline}$. Positive values will indicate improvement after adding the new variables. Cross-validation RMSE will be reported for both configurations to ensure that any gains are not driven by overfitting. In addition, results will be reported with and without HALE_{Birth} to verify that the contribution of the eleven health indicators is not confounded by the inclusion of a near-proxy feature.

IV. IMPLEMENTATION AND PRELIMINARY RESULTS

A. Implementation Environment

All experiments were conducted in Python using Jupyter Notebook. The primary libraries used were Pandas for data manipulation and merging, NumPy for numerical operations, and Scikit-learn for preprocessing, model training, and evaluation. Matplotlib was used for result visualisation. The preprocessing, normalisation, and model training pipelines were implemented across three separate notebooks: data_preprocessing.ipynb for cleaning and imputation, normalized_dataset_export.ipynb for feature engineering and scaling, and model_training_and_evaluation.ipynb for model comparison and analysis. All experiments were run on a standard laptop environment; no GPU acceleration was required given the tabular nature of the dataset and the use of tree-based ensemble methods.

B. Data Preprocessing Pipeline

The raw dataset was filtered to retain only the years 2000 to 2019. Columns with more than 60% missing values were dropped to reduce noise and remove uninformative features. For the year 2000, where no prior data exists for backward filling, missing values were imputed using the earliest available future observation for the same country. The remaining missing values were then filled using linear interpolation applied within each country using grouped operations by Location, estimating intermediate years from surrounding observations. As a final step, any residual gaps were filled using the column-level mean to ensure that no null values remained.

in the dataset. The cleaned data were saved as a new file (cleaned_merged_dataset.csv) for use in subsequent stages.

C. Feature Engineering and Normalisation

Several feature engineering steps were applied before model training. The year variable was transformed into cyclical features using sine and cosine encoding after scaling the year by its maximum value to help models capture temporal patterns more effectively than raw integer values. Categorical variables (Location, ParentLocation, Dim1) were converted to numerical representations using one-hot encoding. HALE at birth (HALE_Birth) was retained as an input feature rather than being dropped, as it captures the overall health profile of a country and provides informative context for predicting HALE at age 60. Although HALE_Birth is conceptually related to the target variable, it is not a direct transformation of HALE_60; rather, it reflects population-wide health conditions from birth onward, whereas HALE_60 specifically measures expected healthy years remaining at age 60. Its inclusion is therefore justified as a complementary predictor, though its high feature importance (see Section IV-D) should be considered when interpreting model results. The target variable, HALE_60, was kept in its original unit (years) and was not scaled, ensuring that model predictions and error metrics remain directly interpretable. The dataset was split temporally rather than randomly: data from 2000 to 2014 were used for training and data from 2015 to 2019 were reserved for testing. This approach preserves chronological order and avoids data leakage from future observations into the training set. Feature scaling was applied after the split to prevent information from the test set from influencing the training process. Numeric features representing rates and exposure levels were standardised using z-score scaling (StandardScaler), while percentage-based features were scaled to the 0–1 range using MinMaxScaler. The scaler was fitted on the training data only and then applied to both training and test sets. The resulting four files (X_train, X_test, y_train, y_test) were exported for model training.

D. Preliminary Results

Three machine learning models were trained and evaluated: Linear Regression as a baseline, Random Forest as a tree-based ensemble with tuned hyperparameters, and Gradient Boosting as an advanced boosting method with optimized hyperparameters. All three models were trained on the extended feature set, comprising the eleven core health indicators along with additional engineered and one-hot encoded features described in Section III-C. To better visualize the differences in model performance, Fig. 2 presents a comparison of R^2 and MAE across the evaluated models.

Gradient Boosting achieved the best performance across all metrics, with a test R^2 of 0.9799, MAE of 0.6126 years, and RMSE of 0.9097 years. The gap between training and test R^2 was small for all three models, indicating no significant overfitting, as confirmed by the small train–test R^2 differences. Feature importance analysis from the Gradient Boosting model identified infant mortality rate (importance = 0.714) as the

TABLE III
PRELIMINARY MODEL PERFORMANCE (EXTENDED FEATURE SET)

Model	Test R^2	Test MAE (years)	Test RMSE (years)
Linear Regression	0.9637	0.7289	1.2220
Random Forest	0.9543	1.0140	1.3716
Gradient Boosting	0.9799	0.6126	0.9097

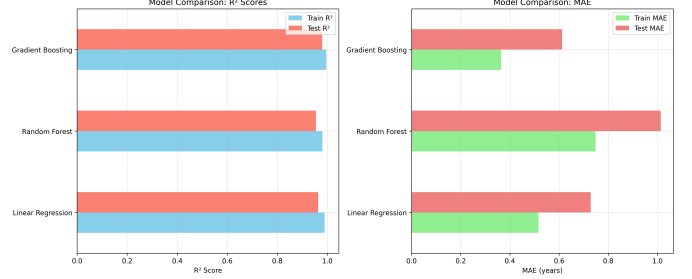


Fig. 2. Comparison of model performance in terms of R^2 and MAE for Linear Regression, Random Forest, and Gradient Boosting.

dominant predictor, followed by HALE at birth (importance = 0.224). Note: the variable labelled “infant mortality rate” in the dataset requires verification against the original predictor list in [7], which uses “adult mortality rate”; this will be confirmed in the final version. The remaining features, in descending order of importance, were probability of dying from chronic diseases, suicide rates, homicide rates, PM2.5 air pollution, obesity prevalence, and physical inactivity. The presence of PM2.5 among the contributing features provides initial support for the inclusion of environmental indicators in the prediction framework. Figure 3 illustrates the relative importance of the top features in the best-performing model.

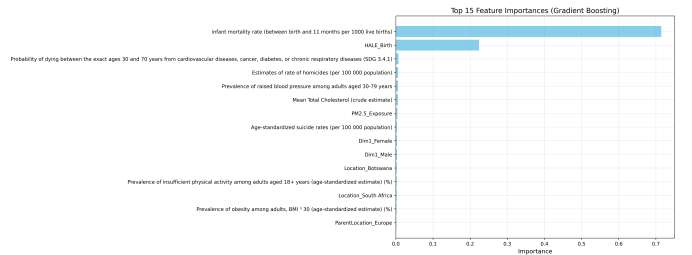


Fig. 3. Top 15 most important features as identified by the Gradient Boosting model.

E. Ablation Study

This section evaluates the contribution of three newly introduced upstream health determinants—PM2.5 exposure, undernourishment, and health expenditure—by comparing model performance under baseline (8 features) and extended (11 features) configurations. Unlike the original study, which employed a random train–test split, the present work adopts a temporal split to better reflect real-world deployment conditions and reduce the risk of temporal leakage [7]. Accordingly,

the analysis focuses on relative performance trends rather than direct comparison of absolute metrics.

TABLE IV
ABLATION STUDY RESULTS (BASELINE VS EXTENDED FEATURE SET)

Model	Configuration	Test R^2	Test RMSE
Linear Regression	Baseline (8 features)	0.9869	1.22
Linear Regression	Extended (11 features)	0.9637	1.49
Random Forest	Baseline (8 features)	0.9886	1.06
Random Forest	Extended (11 features)	0.9543	1.37
Gradient Boosting	Baseline (8 features)	0.9819	1.34
Gradient Boosting	Extended (11 features)	0.9799	0.91

The comparative results are summarized in Table IV. Across both configurations, ensemble methods consistently outperform linear regression, indicating their superior ability to capture complex nonlinear relationships in the data. Under the extended feature set, Gradient Boosting achieves the best performance, with a test R^2 of 0.9799 and an RMSE of 0.9097, demonstrating strong generalization under a temporally constrained evaluation setting.

A notable finding is that the baseline configuration (8 features) yields higher R^2 values than the extended configuration (11 features) for Linear Regression ($\Delta R^2 = -0.0232$) and Random Forest ($\Delta R^2 = -0.0343$). This decline is likely attributable to the sensitivity of these models to increased feature dimensionality. Linear Regression assumes additive, independent relationships among predictors, and the introduction of correlated upstream variables may introduce multicollinearity that degrades coefficient estimates. Random Forest, while more robust to collinearity, can still be affected when additional features dilute the signal from dominant predictors across its bagging process.

In contrast, Gradient Boosting’s sequential residual-correction mechanism is better equipped to selectively incorporate informative features while down-weighting redundant ones, which explains why it is the only model to improve on RMSE ($\Delta \text{RMSE} = +0.43$ years) despite a marginal R^2 decrease ($\Delta R^2 = -0.0020$). These results suggest that the three upstream determinants carry genuine but modest predictive signal that is best captured by boosting methods, and that model selection is a critical factor when expanding feature sets with correlated health indicators.

Overall, while mortality-related variables remain the dominant predictors, the inclusion of upstream determinants enhances the explanatory richness of the model under a more realistic evaluation framework.

F. Issues Encountered and Corrections

Several methodological issues were identified in the initial implementation and corrected during development: (i) the target variable HALE_60 was incorrectly scaled and is now kept in its original unit (Section III-D); (ii) a random train-test split was replaced with a temporal split to prevent look-ahead bias (Section III-E); (iii) feature scaling was moved to after the split to eliminate data leakage (Section III-D); (iv)

RMSE and MAE were added alongside MSE for interpretable error reporting (Section III-E); and (v) the two-tiered median imputation from [7] was replaced with a three-step strategy (backward fill, linear interpolation, column-mean fallback) to better exploit the temporal structure of the data (Section III-D). The feature set was expanded from eight to eleven core health indicators (Section III-C), and the preliminary feature importance results confirm that PM2.5 air pollution contributes to the Gradient Boosting model, providing initial evidence that the inclusion of upstream health determinants adds predictive value.

V. DISCUSSION

A. Key Findings

This study examined whether the integration of upstream health determinants—specifically environmental (PM2.5 exposure), nutritional (undernourishment), and health system (health expenditure) indicators—enhances the prediction of Healthy Life Expectancy at age 60 (HALE_60) [1]. The results indicate that the Gradient Boosting model achieved the highest predictive performance ($R^2 = 0.9799$, RMSE = 0.9097), outperforming both Linear Regression and Random Forest models. Feature importance analysis further reveals that PM2.5 contributes to model performance, although mortality-related indicators remain the most influential predictors.

B. Interpretation of Results

The findings demonstrate that machine learning models, particularly ensemble methods, are effective in capturing complex and non-linear relationships between health indicators and HALE_60. The superior performance of Gradient Boosting can be attributed to its capacity to model interactions among variables more effectively than linear approaches.

The consistently strong performance across models may be explained by the presence of highly informative baseline predictors, such as mortality indicators, which account for a substantial proportion of variability in population health outcomes. This may reduce the observable marginal contribution of newly introduced variables in aggregate performance metrics.

The ablation study reveals that the effect of adding upstream determinants is model-dependent. For Linear Regression and Random Forest, the extended feature set produced lower R^2 and higher RMSE than the baseline, indicating that these models did not benefit from the additional variables under the current experimental conditions. This does not necessarily mean the new variables lack predictive value; rather, it suggests that models with limited capacity to handle correlated or redundant features may be negatively affected by increased dimensionality. Gradient Boosting, by contrast, was the only model to reduce RMSE when moving from 8 to 11 features, demonstrating that the upstream determinants carry complementary information that boosting methods can exploit. These mixed results highlight that feature expansion and model selection must be considered jointly: the value of new variables depends on the model’s ability to integrate them effectively.

Furthermore, the slightly lower absolute R^2 values in the extended configuration compared to the original study should not be interpreted as a limitation of the models. Instead, it reflects a more rigorous evaluation framework, as this study adopts a temporal train-test split that preserves chronological order and better simulates real-world prediction scenarios [7].

C. Comparison to Prior Work

The results are consistent with prior studies, which report that ensemble learning methods outperform linear models in predicting life expectancy-related outcomes [2] [3] [8]. In line with these findings, Gradient Boosting demonstrated superior predictive accuracy, reinforcing the suitability of non-linear modeling approaches for this domain.

This study differs from previous work in two key aspects. First, it focuses on HALE₆₀ rather than general life expectancy, addressing a gap identified in the literature [1]. Second, it incorporates environmental, nutritional, and health system variables that were not considered in earlier models [4] [5] [6].

Additionally, methodological differences must be noted. The original study employed a random train-test split, whereas this study utilizes a temporal split. Random splitting in time-dependent datasets can introduce temporal leakage, potentially leading to higher performance estimates. The temporal validation approach adopted here provides a stricter but more realistic assessment of model generalization, which accounts for the lower absolute metrics observed in this study [7].

D. Implications

The findings have several implications for research and policy. From a research perspective, the results support the integration of upstream determinants into predictive models of population health [1] [11], suggesting that future models should adopt a more comprehensive framework that combines clinical, environmental, and socio-economic factors.

From a policy perspective, the contribution of variables such as PM_{2.5} underscores the importance of environmental health interventions [9]. Improving air quality may contribute not only to reduced disease burden but also to an increase in the number of years lived in good health.

Furthermore, the study highlights the importance of robust evaluation strategies, such as temporal validation, in ensuring the reliability and generalizability of predictive models in health-related applications [7].

E. Limitations

Several limitations should be acknowledged. First, PM_{2.5} data were not consistently available for all countries in the early years of the dataset, particularly around 2000, necessitating interpolation to preserve temporal continuity [4]. Although this approach improved dataset completeness, the use of directly observed data would likely provide a more accurate representation of environmental exposure patterns.

Second, the study evaluated three machine learning models (Linear Regression, Random Forest, and Gradient Boosting).

While these represent a baseline, a tree-based ensemble, and a boosting method, the inclusion of additional algorithms such as Ridge Regression, Lasso, or Support Vector Regression could provide a more comprehensive comparison.

Third, the planned comparison of model performance with and without HALE_{Birth} was not completed. Because HALE_{Birth} accounts for the second-highest feature importance (0.224), its presence may inflate overall R^2 values and partially mask the independent contribution of the three new upstream variables. This remains a priority for future work.

Fourth, the use of country-level aggregated data limits the ability to capture within-country heterogeneity, particularly for environmental and nutritional factors that may vary substantially across regions [10].

F. Future Work

Future research may extend this work in several directions. First, expanding the model set to include additional algorithms such as Ridge Regression, Lasso Regression, AdaBoost, and Support Vector Regression would enable a more comprehensive comparison across both baseline and extended feature sets.

Second, conducting the with-and-without HALE_{Birth} comparison described in Section III-E would clarify how much of the model's predictive power derives from the near-proxy feature versus the eleven core health indicators, and would provide a cleaner assessment of the three new variables' contribution.

Third, the application of deep learning approaches, such as neural networks, may further improve predictive performance by capturing more complex and hierarchical relationships among health indicators, particularly when combined with larger and more granular datasets [3].

VI. CONCLUSION

This study extended the HALE prediction framework proposed by Pranjali et al. [7] by integrating three upstream health determinants—PM_{2.5} air pollution exposure, prevalence of undernourishment, and current health expenditure—into the original feature set of eight socio-economic and clinical indicators. The extended model was evaluated using a temporal train-test split (2000–2014 for training, 2015–2019 for testing), providing a more realistic assessment of predictive performance compared to random splitting.

Among the evaluated models, Gradient Boosting achieved the best performance on the extended feature set, with an R^2 of 0.9799 and RMSE of 0.9097 years. This confirms that ensemble-based approaches are particularly effective in capturing the nonlinear relationships between upstream health determinants and Healthy Life Expectancy at age 60 (HALE₆₀).

Feature importance analysis indicated that PM_{2.5} air pollution contributes meaningfully to prediction performance, supporting the inclusion of environmental indicators in HALE modelling frameworks. However, the ablation study revealed model-dependent effects: while Gradient Boosting benefited from the extended features (RMSE reduction from 1.34 to 0.91 years), Linear Regression and Random Forest showed

slight performance degradation, suggesting that the additional variables provide modest but structured predictive signal that is best captured by boosting-based methods.

Overall, the findings demonstrate that environmental, nutritional, and health-system indicators can be incorporated into HALE prediction frameworks without degrading performance when appropriate models are used. These results align with recommendations from the Global Burden of Disease Study [1], which emphasise the importance of upstream determinants in shaping population health outcomes.

Future work will expand the analysis to additional machine learning algorithms, further investigate the role of HALE_Birth as a predictor, and explore deep learning approaches to better capture complex interactions among health determinants.

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